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Value Problems

A value problem is one where the counted objects have a value. You need to consider the total number of objects and the amount they are worth.

Example 1

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Brad has 20 coins in his pocket, all either quarters or dimes. If he has exactly \$2.90, how many of the coins are quarters?

Step 1

Use a chart.

Let x represent the number of dimes.

If there are x dimes, there must be $20 - x$ quarters.

Now complete the value column. Multiply the number by the value for that coin. Dimes are worth \$0.10 each. Quarters are worth \$0.25 each.

	Number	Value
Dimes	x	$0.1x$
Quarters	$20 - x$	$0.25(20 - x)$

→ how much each set of coins is worth.

$q = \# \text{ of quarters}$

Step 2

Write and solve an equation.

$$0.1x + 0.25(20 - x) = 2.90$$

Use the distributive property.

$$0.1x + 5 - 0.25x = 2.90$$

Combine like terms.

$$5 - 0.15x = 2.90$$

Use inverse operations to isolate the variable.

$$-0.15x = -2.10$$

$$x = 14$$

There are 14 dimes.

→ into $q = 20 - x$
 $= 20 - 14$
 $= 6$

Step 3

Answer the question. How many of the coins are quarters? $20 -$

$14 = 6$ quarters

∴ Brad has 6 quarters.

Not all value problems are about money. As you read the next example, ask yourself, "How can I tell this is a value problem?"

Example 2

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Value

A restaurant has 30 tables. The small tables seat 2 people, and the large tables seat 5 people. When the restaurant is completely filled, there are 117 customers seated. How many small tables are in the restaurant?

Step 1

Use a chart.

Let x represent the number of small tables.

There must be $30 - x$ large tables.

In the value column, multiply the number by how many chairs are at a table of that size.

Tables	Number	Value
Small	x	$2x$
Large	$30 - x$	$5(30 - x)$

Step 2

Write and solve an equation.

$$2x + 5(30 - x) = 117$$
$$2x + 150 - 5x = 117$$
$$-3x = -33$$
$$x = 11$$

Step 3

Answer the question. There are 11 small tables.

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Value Problems



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